Environ Geochem Health (2026) 48:65
https://doi.org/10.1007/s10653-025-02942-7

ORIGINAL PAPER

Arsenic species exposure and urinary metabolic profiling in children and elderly residents near a petrochemical complex in Taiwan

Tzu-Hsuen Yuan · Yi-Ting Hsieh · Chi-Hsin Chen · Chia-Yu Chang · Chang-Chuan Chan

Received: 14 August 2025 / Accepted: 8 December 2025
© The Author(s), under exclusive licence to Springer Nature B.V. 2025

Abstract The petrochemical industry is one of the main anthropogenic sources arsenic exposure. However, there is a lack of studies focusing on inorganic arsenic species and their impact on metabolites. Therefore, the purpose of this study was to investigate the exposure of arsenic species in the urine of residents living near a petrochemical industrial area in central Taiwan to determine whether it is associated with changes in metabolites. We recruited 81 children and 79 elderly residents living near a petrochemical complex in central Taiwan. Study subjects completed questionnaire survey and urine sample were collected to detect four arsenic species by HPLC–ICPMS and metabolites by GCxGC-TOFMS. Compared to the low exposure group, the children had significantly higher concentrations of all arsenic species, while the elderly had only significantly higher concentrations of inorganic arsenic in high exposure group. Specifically, children in the high-exposure area showed about 20-fold higher As^{3+} and tenfold higher As^{5+} concentrations, whereas elderly participants had 1.4-fold higher total arsenic levels. Moreover, high exposure groups performed significantly worse on various arsenic methylation indicators, especially in children. Further, we identified down-regulation of several inorganic arsenic-related methylation pathway metabolites and also found up-regulation of some metabolites of oxidative stress caused related to inorganic arsenic. We concluded that petrochemical industry may contribute to higher inorganic arsenic exposure related to worse arsenic detoxification ability and higher oxidative stress to residents nearby.

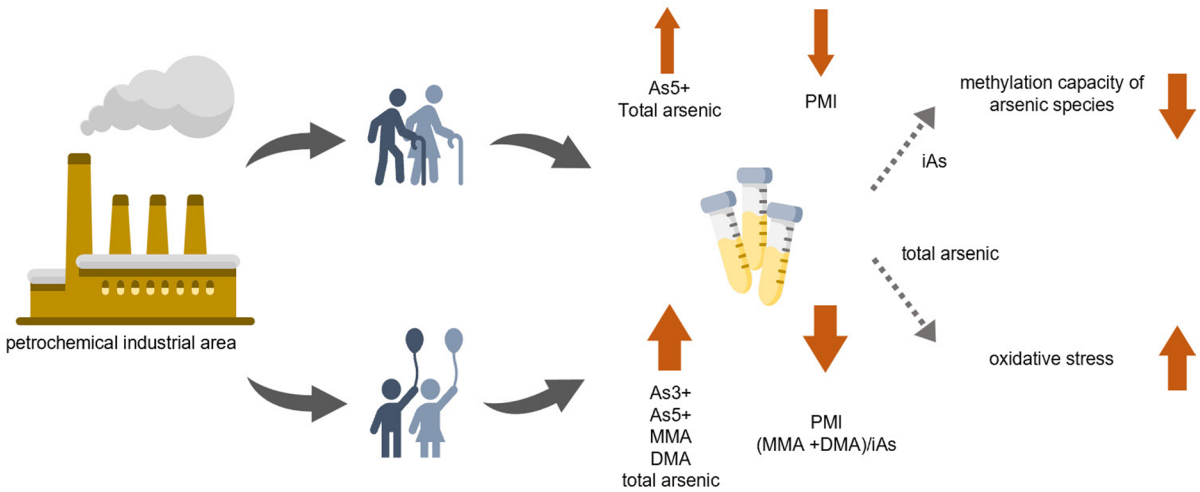
Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10653-025-02942-7.

T.-H. Yuan · C.-Y. Chang
Department of Health and Welfare, University of Taipei, Taipei, Taiwan

Y.-T. Hsieh · C.-H. Chen · C.-C. Chan (✉)
National Taiwan University College of Public Health, Institute of Environmental and Occupational Health Sciences, Taipei, Taiwan
e-mail: ccchan@ntu.edu.tw

Published online: 23 December 2025

Graphical abstract



Keywords Inorganic arsenic · Arsenic methylation · Metabolic profiling · Petrochemical industry

Introduction

Arsenic, a metalloid that occurs naturally, is abundant in the earth's crust (Chang et al., 2019), and its mobility in water and soil is strongly influenced by redox conditions, pH, and interactions with minerals such as iron and manganese oxides (Smedley & Kinniburgh, 2002). It can be divided into different species with diverse physical and chemical characteristics based on their functional groups. Therefore, to accurately reflect the exposure pattern, we cannot simply discuss the level of total arsenic. Arsenic can be classified as organic or inorganic. Arsenobetaine (AsB) and arsenosugar are the main species of common organic arsenic. Arsenobetaine (AsB) is abundant in shellfish or finned fish, while is abundant in seaweed in nature. In humans, inorganic arsenic is metabolized through methylation to form monomethylarsonous acid (MMA) and dimethylarsinous acid (DMA), which are the main urinary metabolites of inorganic arsenic (Taylor et al., 2017). This means that exposure to organic arsenic is related to the consumption of seafood. Inorganic arsenic includes arsenate (As^{5+}) and arsenite (As^{3+}), both of which are mainly used in industrial manufacturing, agriculture, and medicine. However, the exposure hazards of these substances resulting from human behavior are still unclear.

Arsenic produces reactive oxygen species, ROS during its redox cycle and metabolic activation, leading to lipid peroxidation and DNA damage (Ratnaike, 2003). The toxicity of arsenic also differs between organic and inorganic forms. Compared with organic arsenic, inorganic arsenics exhibits much higher toxicity. and can cause both acute and chronic health health effects. Short-term exposure to high levels of inorganic arsenics may result in symptoms such as nausea, vomiting, abdominal pain, and, in severe cases, brain and peripheral nerve diseases (Tyler & Allan, 2014). Chronic exposure to inorganic arsenic is of greater concern, as it can disrupt enzyme activity and normal body functions. Specifically, As^{3+} is more toxic than As^{5+}(Kuo et al., 2017). Overall, the relative toxicity of each arsenic species, from most to least toxic, can be arranged as follows: $As^{3+} > As^{5+} > MMA \approx DMA$ (Petrick et al., 2000).

Due to its higher toxicity, it is crucial to be concerned about the adverse health effects caused by inorganic arsenics. Inorganic arsenics can interfere with cell respiration, glutathione metabolism, and enzyme activity in mitochondria, leading to its accumulation in the body, which may induce skin melanin deposition and keratinization. Furthermore, studies have indicated that inorganic arsenics can disrupt

the normal functioning of insulin, which is associated with type 2 diabetes. Additionally, it is believed that inorganic arsenic can cross the placenta and affect the development of the fetal nervous system. In children, studies have shown that high levels of arsenic are associated with lower scores in memory and language abstraction (Jiang et al., 2014), and arsenic accumulation during childhood may lead to neurobehavioral abnormalities during puberty (Hong et al., 2014). Regarding elderly individuals, studies have found that chronic arsenic poisoning, as well as chronic lead and mercury poisoning, is an independent risk factor for cardiovascular disease (Jennrich, 2013), and there is a significant dose–response relationship between average or cumulative arsenic exposure and the risk of arsenic-related skin lesions (Rahman et al., 2006). Based on previous epidemiology and toxicology studies, the International Agency for Research on Cancer has confirmed that exposure to inorganic arsenic carries the risk of skin, lung, bladder, and other cancers, classifying it as a carcinogen.

Arsenic can enter the body through various pathways such as ingestion, inhalation, and skin absorption. It is primarily metabolized by the liver and converted into different types of arsenic species. Organic arsenic is less easily absorbed by cells, whereas inorganic arsenic readily binds to red blood cells and accumulates in the liver, kidneys, muscles, bones, hair, skin, and nails (Hong et al., 2014). Among these species, the ingested arsenic takes an average of 3 to 4 days to be processed, and approximately 50% of the arsenic is excreted through urine. The detectable arsenic species in humans include DMA, MMA, As^{3+}, and As^{5+}. Upon entering the human body, As^{5+} is reduced to As^{3+} by glutathione (GSH), and As^{3+} is further methylated to form MMA^{5+}. Subsequently, MMA^{5+} is reduced to MMA^{3+} by GSH, and MMA^{3+} is further methylated to produce DMA^{5+}. Finally, DMA^{5+} is reduced to DMA^{3+}, which is considered the final form for excretion from the body. The main species in urinary arsenic are 60–80% DMA, approximately 10 - 20% MMA, and around 10 - 20% As^{5+} and As^{3+}, although individual differences in metabolic efficiency exist (Gamble et al., 2005). In recent years, studies have indicated that environmental exposure plays a significant role in arsenic metabolism (Giri & Dey, 2017; Hall et al., 2013). A study conducted in China found that the level of GSH is associated with the pathway of arsenic methylation and the oxidative stress caused by inorganic arsenic in chronic arsenic-exposed environments (Li et al., 2017). However, research in this area is still limited, and further evidence is needed to clarify the relationship between environmental exposure and metabolomics.

In recent years, industrial activities have emerged as significant pollution sources, and it has been demonstrated that petrochemical industries can emit various heavy metals, including inorganic arsenic. Inorganic arsenic can be deposited in soil and water sources through atmospheric sedimentation, wastewater discharge, and inadequate waste treatment. (Patel et al., 2005). A previous study found that arsenic concentrations in the atmosphere are higher in areas near petrochemical industries (Bosco et al., 2005). This suggests a strong correlation between arsenic emissions and petrochemical industries. Furthermore, our previous studies conducted in the largest petrochemical complex in Taiwan have demonstrated that residents living near the petrochemical complex have higher levels of urinary heavy metals, including vanadium, arsenic, strontium, and manganese, compared to residents living 10 km away from the complex (Chen et al., 2017).

With respect to environmental exposure, the World Health Organization (WHO), the guideline value for arsenic in drinking water is 10 μg/L (WHO, 2022), based on extensive evidence linking chronic arsenic exposure to cancer and other adverse health outcomes. Based on the Environmental Epidemiology Study in Coastal Areas of Yunlin County, water samples collected within a 10-km radius of the sixth naphtha cracker complex (No. 6 NCK) showed an arsenic concentration of 1.12 ± 1.77 ng/mL. Although below the WHO guideline, this level is approximately three times higher than concentrations measured at distances of 10 - 20 km from the complex (0.34 ng/mL), indicating that arsenic levels in nearby communities may still exceed regional background values (Chen et al., 2013). Regarding air pollution, Yunlin air-quality monitoring data showed that during winter periods when air quality deteriorated, the average arsenic concentration in PM_{10} reached 6.67 ng/m^3, exceeding the European Union's target value of 6 ng/m^3 for total arsenic in particulate matter (Chen et al., 2015). Based on the Environmental Epidemiology Study in Coastal Areas of Yunlin County, arsenic concentrations in particulate matter near the industrial area were significantly higher than in

non-affected areas when the wind direction was influenced by emissions from the petrochemical complex. (Chan et al., 2012, 2015).

Children and older adults, the primary vulnerable groups examined in this study, may be particularly affected by such exposure. Children exhibit higher susceptibility due to immature physiological development and higher respiratory rates, while older adults experience age-related declines in immune function. Previous research conducted in vicinity of the same petrochemical complex has reported higher rates of respiratory diseases and subclinical liver impairment among children living near the petrochemical complex, as well as significantly elevated cancer incidence and mortality among older adults in high-exposure zones (Wang et al., 2019; Weng et al., 2008; Yuan et al., 2018). These findings highlight the potential vulnerability of these populations to arsenic exposure emitted from petrochemical activities.

To discuss the issue of occupation safety, numerous studies have explored the relationship between arsenic exposure and workers (Byun et al., 2013; Yamauchi et al., 1989). However, there are still limited studies that have sought to understand arsenic exposure among residents living near petrochemical industries. Furthermore, most previous studies have primarily relied on total arsenic measurements to assess exposure status (Lasky et al., 2004; Navas-Acien et al., 2011). However, based on the available evidence, the toxicity of arsenic can vary significantly depending on the specific arsenic species. To provide a clearer understanding of the health effects by arsenic exposure from petrochemical industries, our study examined the arsenic exposure among children and the elderly living near a petrochemical complex. Specifically, we aim to investigate the concentrations and distributions of various arsenic species in urinary samples of residents nearby, examine the differences in methylation efficiency of arsenic species among residents, and characterize the expression of metabolites along the arsenic methylation pathway.

Materials and methods

Study area and subjects

Our study was conducted in Yunlin County, specifically at the No. 6 Naphtha Cracking Complex located in Mailiao Township. This complex, spanning an area of approximately 2603 hectares, is considered a petrochemical industrial complex in Taiwan. It began operations in 1992 and consists of various facilities, including a thermal power plant capable of generating 1.8 million kW of electricity, three cogeneration plants producing 2.82 million kW of electricity, three oil refineries with a daily oil production of 540,000 barrels, and an annual production capacity of 2.9 million tons of ethylene. Additionally, there are two naphtha cracking plants and twelve aromatic and plastic plants within the complex.

The study area comprised ten townships with a similar socioeconomic status in Yunlin County, located in central Taiwan. We employed administrative districts to divide the participants into high and low exposure groups based on their proximity to the complex. Specifically, high exposure areas were defined as administrative districts located within approximately 10 km of the petrochemical complex, whereas low exposure areas were located beyond 10 km (Fig. 1). Previous studies have demonstrated that this distance-based classification reflects significant differences in petrochemical emission concentrations (Yuan et al., 2016, 2020).In addition to geographical classification, biomarkers of petrochemical industrial emissions were also used as exposure indicators. These biomarkers have been validated as reliable markers of petrochemical-related exposure in this region (Chen et al., 2019). Among participants residing in high-exposure administrative districts, those whose urinary concentrations of petrochemical related biomarkers were within the top 60% of the cohort were classified as the high-exposure group. Conversely, among participants living in low exposure districts, those whose biomarker concentrations fell within the bottom 40% were assigned to the low exposure group. The high and low exposure groups in this study were therefore defined using this combined criterion. The study subjects were recruited between 2009 and 2012 and completed both a questionnaire survey and urine sample collection. We first established a cohort of 3,230 residents who had lived at the same address within the study area for more than 5 years and were either elderly adults (> 55 years) or children (9–15 years). From this cohort, a total of 160 study subjects were finally included in the analysis. The study subjects were further divided into four groups based on their administrative district and age, which included elderly individuals at high risk of

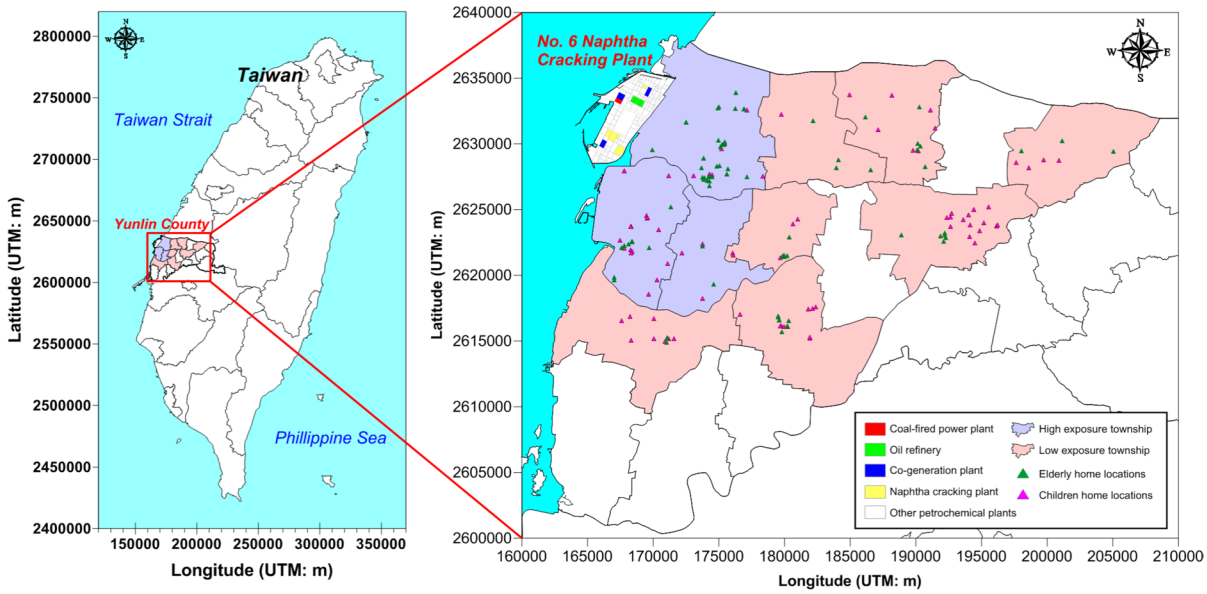


Fig. 1 The locations of the petrochemical complex and the study subjects from Yunlin County in central Taiwan

exposure (n = 44), children at high risk of exposure (n = 30), elderly individuals at low risk of exposure (n = 35), and children at low risk of exposure (n = 51).

Questionnaire and health data

The present study involved a questionnaire investigation and health examination for each participant. Well-trained interviewers collected personal information from the study subjects, including basic demographic characteristics, medical history, personal behaviors, lifestyle habits, dietary information, and information about their home environment. The height and weight of each study subject were measured during the health examination. Additionally, first morning spot urine samples were collected from each participant. The urine sample was collected in a urine collection bag and transferred to a centrifuge tube, which was then stored in a refrigerator at -20 °C. This study received approval from the Research Ethics Committee of the National Health Research Institutes in Taiwan (approval number: 201505016RINB), and written informed consent forms were obtained from each participant.

Arsenic speciation analysis

The present study analyzed the urinary levels of As^{3+}, As^{5+}, DMA, and MMA. These arsenic species levels were determined using high-performance liquid chromatography coupled to inductively coupled plasma-mass spectrometry (HPLC–ICPMS) with an Agilent 1100 system and an Agilent 7500 ICP-MS instrument. The standard solution of As^{3+} used in the analysis was purchased from High-Purity Standards (Arsenite, 1000 µg/mL in 2% HCl, High-Purity Standards, USA). The standard solution of As^{5+} was obtained from Merck (Arsenic ICP standard, 1000 µg/mL in 2–3% HNO3, Merck, USA). The standard solution of DMA was purchased from ChemService (DMA solution 1000 µg/mL in water, ChemService, USA). Finally, the standard solution of MMA was obtained from ChemService as well (MMA solution 100 µg/mL in water, ChemService, USA). For the HPLC-ICP-MS method, analytical calibration standards of mercury species were prepared daily in the mobile phase. The calibration standards covered a concentration range of 0.1 - 500 µg/L, and suitable serial dilutions of the stock solution were made to prepare these standards. The urine samples were removed from the -20 °C refrigerator and allowed to thaw. They were then homogenized using a

shaker. The urine samples were transferred to centrifuge tubes and diluted three times with distilled water (Millipore, Milli-Q Elix®5/RiOs). After dilution, the prepared samples were filtered using a 0.45 µm polyvinylidene fluoride syringe filter.

The mobile phase used in the analysis consisted of 5 mM tetrabutylammonium hydroxide (20% in water, Merck, USA), 3 mM malonic acid (Sigma-Aldrich, USA), and 5% methanol (Merck, USA). A 50µL injection volume of the sample was used, and the gradient flow rate was set at 1.5 ml/min. The ICP-MS conditions were set as follows: RF power at 1500W, sampling depth at 7 mm, torch-H at 0.8 mm, torch-V at 0.75 mm, carrier gas at 1.15 L/min, makeup gas at 1.00 L/min, helium gas at 2.00 L/min, and nebulizer pump at 0.3 rps. To ensure accurate and precise measurements, certain standards and criteria were followed. The coefficient of determination (R2) of the calibration curve was required to be higher than 0.995. The relative error of the spiked samples in each batch was defined as within 15%. The recovery rate was determined to be in the range of 80% to 120%. In addition, the coefficient of variation (CV) for repeated measurements of urinary arsenic species was 4.0%. The method detection limits (MDL) for As^{3+}, As^{5+}, MMA, and DMA were determined to be 0.18 µg/L, 0.20 µg/L, 0.24 µg/L, and 0.32µg/L, respectively. For samples that were below the detection limit, 1/2 MDL was used to represent the arsenic species level. The proportions of samples below the detection limit were 0.06% for As^{3+}, 23.13% for As^{5+}, 0% for DMA, and 0.25% for MMA.

Metabolomics analysis

The database of urine metabolites is a previously published study (Chen et al., 2017). Urine metabolite profiling in this study followed the analytical workflow described in Chen et al. (2017), with detailed sample preparation procedures reiterated here for clarity. Urine samples were collected using sterile containers and stored on ice or at - 80 °C to preserve metabolite stability. A fixed volume of each sample was aliquoted into microcentrifuge tubes for consistent handling. Urease was added to remove urea and reduce interference during GC × GC-TOFMS analysis, followed by the addition of two internal standards (2-chlorophenylalanine and heptadecanoic acid) for normalization and quality control. Proteins were precipitated by adding cold methanol, and the mixture was centrifuged to obtain the supernatant.

A two-step derivatization procedure was then performed, consistent with the protocol of Chen et al. (2017). Briefly, methoxyamination was carried out using methoxyamine hydrochloride in pyridine at 60 °C for 2 h, followed by silylation with BSTFA containing 1% TMCS at 70 °C for 1 h to increase metabolite volatility and thermal stability. The derivatized samples were subsequently analyzed using a Pegasus 4D GC × GC-TOFMS system for comprehensive separation and detection.

For quality control (QC), blanks and pooled QC samples were analyzed at the beginning of each batch, and every five samples within batch. After data cleaning and preprocessing, acquired data were outputted as peak area for NIST library (version 08, National Institute of Standards and Technology, Gaithersburg, MD, USA) identified potential metabolite peaks, and we selected those with mass spectrum matched library spectrum for > 60% (similarity score > 600 out of 1000). Obtained potential metabolite features were preprocessed by removing those with > 50% missing values, and replacing the missing values of the remaining features with half of the minimum positive value in the original data. Preprocessed data were normalized by sum of total peak area, log transformed, and autoscaled (mean-centered and divided by the standard deviation of each variable) prior to further statistical analysis. NIST library match showed the derivatized form of potential metabolite peaks, which we converted to underivatized forms by replacing methoxyamine and trimethylsilyl groups with carbonyl and hydroxyl functional groups, respectively. ChemSpider was used for the identification of the underivatized forms of potential metabolite peaks (Royal Society of Chemistry, London, UK). All potential metabolites peaks were then put through an online repository (http://cts.fiehnlab.ucdavis.edu/) and searched under three online databases: the Human Metabolome Database (HMDB) (Wishart et al. 2018), Kyoto Encyclopedia of Genes and Genomes (KEGG) (Kanehisa et al. 2014), and Chemical Entities of Biological Interest (ChEBI) (Hastings et al. 2012), for identification of known metabolites, chemical class, and involved biological pathways.

Statistical analysis

The comparisons of basic characteristics between the study subjects living in the HE area and those in the LE area were performed using an ANOVA test for continuous variables and chi-squared tests for categorical variables. Fisher's Exact test was used when the expected frequency was less than 5. The urinary arsenic levels were adjusted by urine creatinine, and if the levels were below the MDL, they were replaced by 1/2 MDL. Total arsenic was calculated by summing up the levels of As^{3+}, As^{5+}, DMA, and MMA. To understand the methylation degree of arsenic species, the primary methylation index (PMI), secondary methylation index (SMI), and the ratio of organic arsenic (OrgAs) to inorganic arsenic (iAs) were utilized to explore the relationship between arsenic exposure and metabolism. PSI was calculated as MMA/iAs, and SMI was calculated as DMA/MMA. Regression models were used to compare the urinary levels of arsenic species, PMI, SMI, and (MMA + DMA/iAs) in children and the elderly, adjusting for factors such as gender, BMI, drinking habits, betel nut consumption, drinking water sources, fish consumption, and education level. The analysis was performed using SAS 9.4 for Windows, and the significance level was set at 0.05.

The metabolite data was analyzed using the online software MetaboAnalyst 4.0 (The Metabolomics Innovation Center, Edmonton, Alberta, Canada, http://www.metaboanalyst.ca/). For identifying potential metabolite peaks related to exposures, the peak area data were normalized by the sum of the total peak area, ensuring that the cumulative peak area summed up to 1 in each subject. The data was then log-transformed and autoscaled before conducting partial least squares discrimination analysis (PLS-DA) between the high and low concentration groups. The PLS-DA models were validated using permutation tests and cross-validation tests. The variable importance in the projection (VIP) values of the components were averaged, and metabolites with VIP > 1 and FDR < 0.05 after variance analysis were analyzed using a heatmap. The heatmap employed Euclidean two-point distance measurement and the Ward method to calculate the distance between groups, aiming to compare the differences in metabolite performance between the groups. PLS-DA provides two values to represent its model evaluation parameters: R^2 and Q^2. R^2 signifies the model's ability to explain the Y variable, while Q^2 represents the model's predictive ability in judgment.

Results

Table 1 shows that there were no significant differences in gender and body mass index (BMI) between the children and elderly groups, as well as the high and low exposure groups. The low-exposure group has a higher number of obese people, but there is no statistically significant difference between the two groups. The elderly group has a similar distribution in the high and low exposure groups, with more than half of the individuals classified as obese. Regarding drinking habits, a small number of individuals in the child group reported drinking, but the distribution was similar between the high and low exposure areas, and there was no statistically significant difference. Similarly, the elderly group showed a similar distribution in the high and low exposure groups. Similarly, the elderly group showed a similar distribution in the high and low exposure groups. As for educational attainment, it was divided into three groups: elementary school, junior high school to high school, and university or above. There were significant differences between the high and low exposure groups in both the children and elderly groups. In the high-exposure group of the child group, three individuals had an educational attainment of elementary school, whereas in the low-exposure group, all individuals had at least a junior high school or senior high school education. In the elderly group, approximately 85.4% of individuals in the high-exposure group had an educational attainment below elementary school, while the proportion of individuals with an educational attainment above junior high school was relatively higher in the low-exposure group. There were significant differences between the high and low exposure groups in terms of educational attainment. Regarding smoking habits, it was observed that some individuals in the child group reported smoking, but there was no significant difference between the high and low exposure groups. The same pattern was observed in the elderly group. In terms of dietary habits before admission, no significant differences were found between the high and low exposure groups, regardless of whether they were in the elderly or child group. In the regression

Table 1 Demographic characteristics of the study subjects living in the high exposure (HE) and low exposure (LE) areas in Yunlin County, Taiwan near the petrochemical complex[a]

	Children (n = 81)			Elderly (n = 79)		
	HE (n = 30)	LE (n = 51)	*p*	HE (n = 44)	LE (n = 35)	*p*
Sex			0.3389			0.6328
Male	18	25		19	17	
Female	12	26		25	18	
BMI[b]			0.6189			0.9891
Underweight	6	12		0	0	
Normal	18	25		10	8	
Overweight	6	14		34	27	
Education Level[c]			0.0214*			0.0089*
Below primary school	3 (10)	0 (0)		35 (85.37)	18 (52.94)	
Middle to High school	27 (90)	51 (100)		5 (12.2)	13 (38.24)	
College	0 (0)	0 (0)		1 (2.44)	3 (8.82)	
Work exposure[c]			0.4403			0.3422
No	30(100)	50(98.04)		39(97.14)	34(88.64)	
Yes	0(0)	1(1.96)		4(2.86)	1(9.09)	
Drinking[c]			0.2788			0.4704
No	28 (93.33)	50 (98.04)		40 (90.91)	30 (85.71)	
Yes	2 (6.67)	1 (1.96)		4 (9.09)	5 (14.29)	
Betel Nut Chewing[c]			0.1895			0.8413
No	29(96.67)	51(100)		41(93.18)	33(94.29)	
Yes	1(3.33)	0(0)		3(6.82)	2(5.71)	
Smoking[c]			0.4944			0.3727
No	27 (90)	48 (94.12)		43 (97.73)	31 (88.57)	
Yes	3 (10)	3 (5.88)		1 (2.27)	4 (11.43)	
Fish intake[c]			0.4313			0.4017
No	18 (60)	26 (50.98)		35 (79.55)	25 (71.43)	
Yes	12 (40)	25 (49.02)		9 (20.45)	10 (28.57)	
Water Source[c]			0.7913			0.0801
Tap water	22 (73.33)	36 (70.59)		41 (93.18)	28 (80)	
Other	8 (26.67)	15 (29.41)		3 (6.82)	7 (20)	

[a]Continuous variables were used ANOVA test, and category variables were analyzed using Chi-square Fisher's Exact test was used when the expected frequency was less than 5.[b] BMI = kg/m^2;[c] Unit: %

models, education level was included as an adjustment variable, and no clear association was observed between education level and either arsenic exposure or arsenic methylation capacity (data not shown). In addition, because the distribution of education levels in our study population was highly unbalanced, stratified analyses by education level were not suitable for evaluating its potential modifying effect on arsenic exposure and metabolism.

Table 2 shows the geometric mean concentrations of urinary arsenic species and total arsenic among children and elderly participants in the high- exposure and low-exposure groups. A total of 81 children samples were analyzed, with 30 samples from the high-exposure area and 51 samples from the low-exposure area. Additionally, 79 samples from the elderly were analyzed, including 44 samples from the high-exposure area and 35 samples from the low-exposure area. This value has been corrected by using the Creatinine value in urine. Comparing the total arsenic concentrations between children and the elderly, the data reveals that the total arsenic concentration in the high-exposure group of children is 91.83 µg/g-creatinine, which is higher than the total arsenic concentration in the high-exposure group of the elderly, which is 86.92 µg/g-creatinine. In the low-exposure area, the total arsenic concentration in the urine of the elderly is 52.29 µg/g-creatinine, whereas in the low-exposure area of children, it is 31.36 µg/g-creatinine. In the children group, after controlling for related variables,

Table 2 Geometric means (GM ± GSD) and 95% confidence intervals of urinary arsenic species among the study subjects living near the petrochemical complex[a]

	Children (n = 81)					Elderly (n = 79)				
	HE (n = 30)	95%CI	LE (n = 51)	95%CI	*p*	HE (n = 44)	95%CI	LE (n = 35)	95%CI	*p*
As^{3+}	26.57 ± 3.33	(17.27–40.89)	4.57 ± 4.28	(3.07–6.82)	< 0.001*	23.06 ± 1.23	(21.72–24.49)	11.75 ± 1.28	(10.83–12.75)	0.214
As^{5+}	1.14 ± 3.91	(0.70–1.85)	0.29 ± 5.46	(0.18–0.46)	< 0.001 *	1.09 ± 1.30	(1.01–1.18)	0.34 ± 1.36	(0.31–0.38)	0.006*
MMA[b]	6.89 ± 3.57	(4.37–10.86)	1.78 ± 3.63	(1.25–2.54)	< 0.001*	6.79 ± 1.16	(6.50–7.08)	5.70 ± 1.27	(5.27–6.16)	0.550
DMA[c]	41.60 ± 3.16	(27.55–62.81)	18.19 ± 3.31	(13.09–25.28)	0.003 *	35.93 ± 1.19	(34.17–37.79)	24.34 ± 1.25	(22.63–26.19)	0.172
Total[d]	91.83 ± 2.65	(64.74–130.27)	31.36 ± 2.53	(24.31–40.47)	< 0.001*	86.92 ± 1.13	(83.82–90.1)	52.29 ± 1.20	(49.24–55.52)	0.006*

[a]Regression model is adjusting for gender, BMI, drinking habits, betel nut, drinking water sources, fish consumption, education level, *$p < 0.05$

[b]MMA = Monomethylarsonic acid (Unit: μg/g-creatinine)

[c]DMA = Dimethylarsinic acid (Unit: μg/g-creatinine)

[d]Total = $As^{3+} + As^{5+}$ + MMA + DMA (Unit: μg/g-creatinine)

the As^{3+} and As^{5+} concentrations in the high exposure groups 26.57 μg/g-creatinine and 1.14 μg/g-creatinine, low exposure groups were 4.57 μg/g-creatinine and 0.29 μg/g-creatinine respectively. Both As^{3+} and As^{5+} concentrations are higher in the high-exposure area. However, when comparing organic arsenic, the concentrations of MMA and DMA in the high-exposure group are significantly higher than those in the low-exposure group, and there are statistically significant differences. As for the elderly group, only the concentrations of As^{5+} and total arsenic were significantly higher in the high-exposure group compared to the low-exposure group, with values of 1.09 μg/g-creatinine vs 0.34 μg/g-creatinine and 86.92 μg/g-creatinine vs 52.29 μg/g-creatinine, respectively. The concentration differences of As^{3+}, MMA and DMA did not reach statistical significance. Compare the percentage of arsenic species in the high and low exposure groups, the percentage of As^{3+} in the high-exposure group is higher than that in the low-exposure group, showing that the high-exposure group has a higher exposure level, while the percentage of MMA is lower than that of the exposure group, and it is statistically significant difference.

To examine the percentage distribution of arsenic species concentrations in the body, a percentage bar graph was made to show the concentration of each arsenic species in the urine of children and the elderly in the high and low exposure groups, as shown in Fig. 2. The bar graph reveals that the primary arsenic species in the urine of both children and the elderly is predominantly DMA, followed by As^{3+}, and with the lowest percentage being As^{5+}. After controlling for related variables, a comparison between the high and low exposure groups in the child group demonstrates that As^{3+} and iAs concentrations are higher in the high exposure group compared to the low exposure group. On the other hand, DMA and OrgAs concentrations are higher in the low exposure group. These differences are statistically significant. Regarding the high-exposure group of the elderly, As^{3+} concentrations are significantly higher than those of the low-exposure group. However, the MMA concentration in the low-exposure group is significantly higher than that in the high-exposure group, while the remaining arsenic species do not exhibit statistically significant differences.

In Table 3, we use PMI, SMI and (DMA + MMA)/iAs as indicators to evaluate the degree of methylation

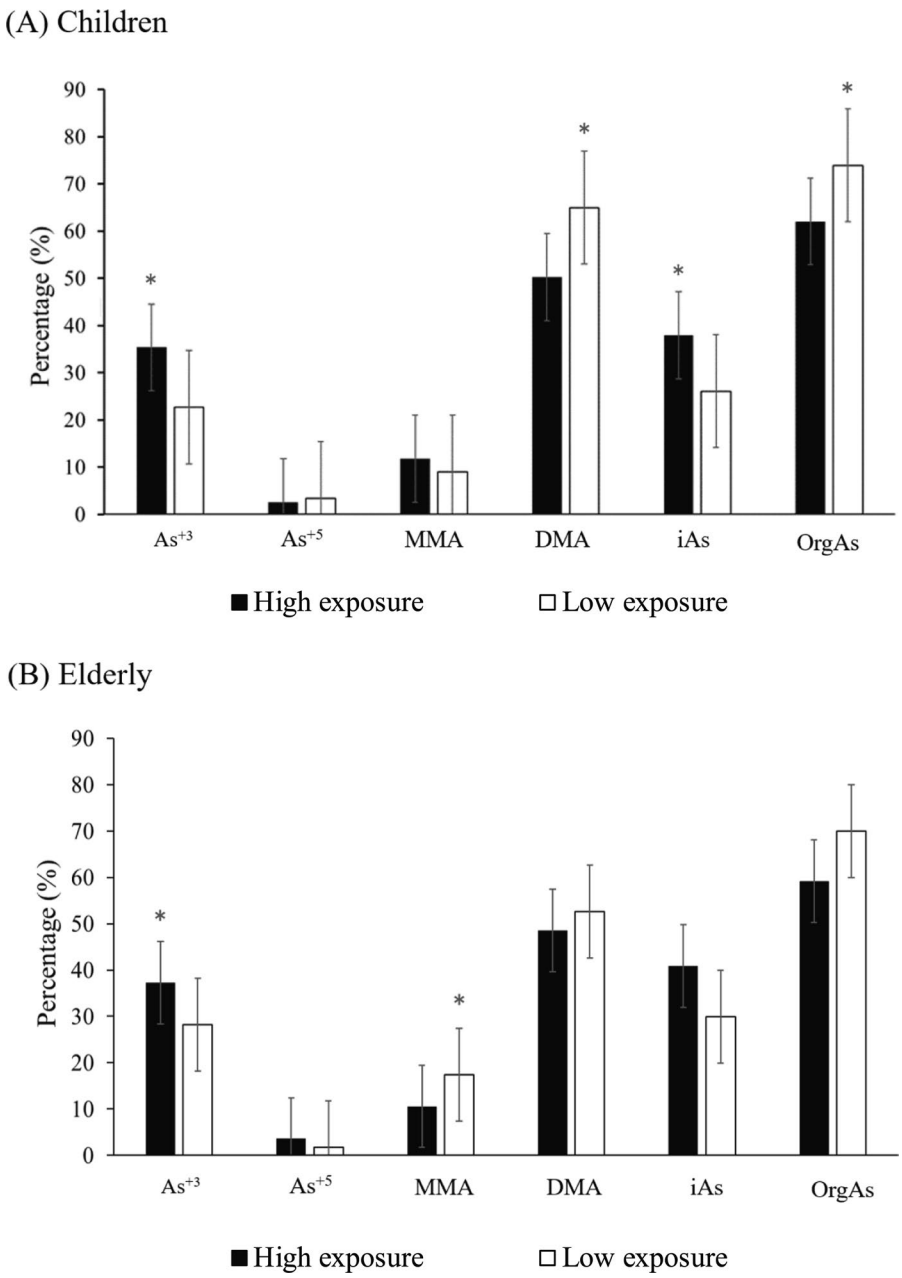


Fig. 2 The percentage of different urinary arsenic species among the study subjects living near the petrochemical complex[a]. [a]Regression model is adjusting for gender, BMI, drinking, smoking, betel nut, water sources, fish consumption, education level, $*p < 0.05$

metabolism of inorganic arsenic in the body. A higher value indicates better methylation metabolism, and these parameters are often used to assess conversion efficiency of arsenic species or as a basis for disease diagnosis. After controlling for related variables in the child groups, the PMI exposure levels of the high and low exposure groups were 0.61 ± 0.20 and 0.75 ± 0.19 respectively. There was a significant difference between two groups, indicating differences in the conversion of inorganic arsenic to MMA. However, when comparing the SMI values between the high and low exposure groups, there was no significant difference, suggesting that the conversion of MMA to DMA did not differ between the two groups. As for the ratio of organic arsenic (MMA and DMA) to inorganic arsenic (As^{3+} and As^{5+}), the high exposure group had a significantly lower value compared to the low exposure group (2.69 ± 1.55 vs

Table 3 The levels of urinary PMI, SMI, (MMA + DMA)/iAs among the study subjects living near the petrochemical complex[a]

	Children (n = 81)			Elderly (n = 79)		
	HE (n = 30)	LE (n = 51)	*p*	HE (n = 44)	LE (n = 35)	*p*
PMI[b]	0.61 ± 0.20	0.75 ± 0.19	**0.002***	0.63 ± 0.15	0.73 ± 0.17	**0.008***
SMI[c]	0.87 ± 0.08	0.87 ± 0.11	0.845	0.75 ± 0.19	0.82 ± 0.17	0.127
(MMA + DMA)/iAs[d]	2.69 ± 1.55	8.44 ± 5.41	**0.014***	4.92 ± 6.28	5.09 ± 4.66	0.749

[a]Regression model is adjusting for gender, BMI, drinking, smoking, betel nut, water sources, fish consumption, education level, $*p < 0.05$

[b]PMI: primary methylation index; PMI = DMA + MMA / (DMA + MMA + As^{3+} + As^{5+})

[c]SMI: secondary methylation index; SMI = DMA / MMA + DMA

[d]MMA: monomethylarsonic acid; DMA: dimethylarsinic acid; iAs: (As^{3+} + As^{5+})

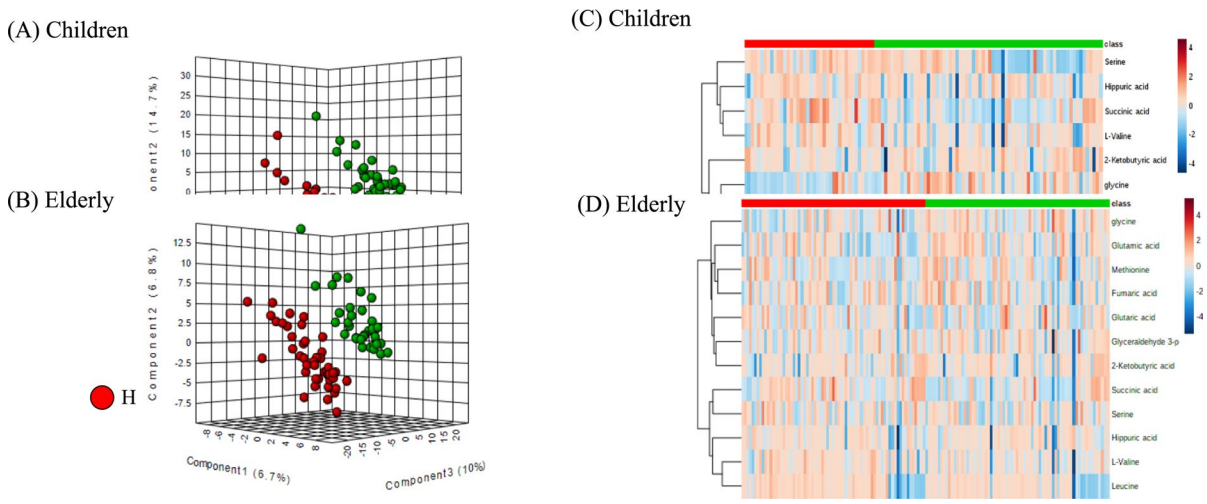


Fig. 3 Score plots of urinary metabolite peaks of PLS-DA from **A** children ($R^2 = 0.66$, $Q^2 = 0.38$) and **B** elderly study subjects ($R^2 = 0.77$, $Q^2 = 0.15$); Heat map of urinary metabolites levels in **C** children and **D** elderly study subjects

8.44 ± 5.41). This indicates that the degree of conversion of inorganic arsenic into organic arsenic by the methylation process in the high-exposure group was lower than that in the low-exposure group. In terms of the comparison between the high and low exposure groups in the elderly, the results showed that after adjusting for related variables, the PMI of the high-exposure group was 0.63 ± 0.15, while it was 0.73 ± 0.17 in the low-exposure group. There was a significant difference, indicating that there was a difference in the conversion of inorganic arsenic to MMA between two groups. However, there were no statistically significant difference in SMI and (MMA + DMA)/iAs between the two groups.

The PLS-DA identification results of metabolites are shown in Fig. 2 and Fig. 3(A). The distinction between the high-exposure group and the low-exposure group is evident for both children and the elderly. The evaluation parameters of the child group are R2 = 0.66, Q2 = 0.38, and that of the elderly group are R2 = 0.77, Q2 = 0.15. Although the PLS-DA effectively distinguishes the metabolites between the high-exposure and low-exposure groups, the Q2 values are slightly lower, indicating relatively poor predictive ability. The difference in model performance between the two age groups can be partly explained by their physiological characteristics. This may be attributed to the high heterogeneity in the elderly population, including chronic diseases, polypharmacy, and variations in metabolic degradation, which introduce substantial background noise into the model (Lawton et al., 2008). To understand the

important metabolites related to arsenic metabolism, we selected variables with VIP (Variable Importance in Projection) values greater than 1, which represents the overall contribution of the variable to the entire model. The threshold of VIP is typically set to greater than 1. In the children and elderly groups, there are 35 and 42 metabolites, respectively, that meet this criterion. In addition, after controlling the related variables, there are still 11 metabolites and 12 metabolites respectively have significant differences between groups. A Heat Map was used to visualize the differences in metabolite expression between the high and low exposure groups, as shown in Fig. 3(C) and (D). The horizontal axis at the top of the graph represents the high exposure group (in red) and the low exposure group (in green). The vertical axis on the right represents the selected metabolites of important influence. Dark brown color indicates up-regulation, while dark blue color indicates down-regulation. Therefore, it can be found from the Heat Map that the following metabolites were relatively down-regulated in the high-exposure group compared to the low-exposure group in children: Glyceraldehyde-3-phosphate, Glutamic acid, Methionine, Glutaric acid, Fumaric acid, and Glycine. On the other hand, the relatively up-regulated metabolites were: 2-ketobutyric acid, L-Valine, Succinic acid, Hippuric acid, and Serine. Similarly, in the elderly group, the down-regulated metabolites in the high-exposure group compared to the low-exposure group were: Glyceraldehyde-3-phosphate, Glutaric acid, Fumaric acid, Methionine, Glutamic acid, and Glycine. The up-regulated metabolites were: Leucine, L-Valine, Huppuric acid, Serine, Succinic acid, and 2-ketobutyric acid.

Discussion

In this study, the study subjects were divided into a low-exposure group and a high-exposure group based on their distance from the petrochemical complex for comparison. The study showed significant differences between the high-exposure and low-exposure groups in both the child and elderly groups. In the child group, there were significant differences in the concentrations of organic arsenic (MMA, DMA) and inorganic arsenic (As^{3+}, As^{5+}) between the high-exposure and low-exposure areas. On the other hand, in the elderly group, although only the concentration of the As5+had a significant difference between high and low exposure, the percentages of As^{3+} and DMA in the high-exposure area indicated a higher exposure level of these arsenic species. Compared to the previous study (Hsueh et al., 2009), the total arsenic content of our research is much higher than that of normal healthy people (20.71 ± 1.10 μg/g-creatinine) whether in the high-exposure group (children: 139.1 μg/g-creatinine; elderly: 122.15 μg/g-creatinine) or the low-exposure group (children: 47.68 μg/g-creatinine; elderly: 90.05 μg/g-creatinine). Additionally, when examining the percentage of inorganic arsenic, in Hsueh's study, the percentage of inorganic arsenic in the body of normal healthy people is only 6.67 ± 0.62% of the total arsenic, even if the percentage of inorganic arsenic in the body of CKD patients is only 7.50 ± 1.04%. Even in the low exposure group of our study, the proportion of inorganic arsenic remained substantially higher than that of healthy populations (children: 26.07%; elderly: 29.98%), while the high exposure group showed the greater proportions (children: 37.96%; elderly: 40.83%), reaching four- to six-fold higher than those reported for healthy individuals. These quantitative findings indicate that residents in the study area are exposed to considerably elevated levels of inorganic arsenic, and that the degree of exposure was potentially associated with proximity to the petrochemical complex. Regarding the differences of the proportion of inorganic arsenic in the body between our study subjects and the normal population in previous studies, a study conducted in Taiwan in 2007 suggested that the increase in inorganic arsenic exposure might be related to living in industrial areas, male gender, and the use of groundwater (Wang et al., 2007). However, in our study, gender and drinking water did not show the statistically significant differences between the high and low exposure groups. The only variable is the distance from the petrochemical complex. Therefore, we can infer that these differences are attributed to the impact of the petrochemical complex.

Arsenic pollution in the vicinity of heavy industry areas has been widely discussed, which can be seen in different environmental media, such as air, water, and soil. Numerous international studies have focused on environmental testing and human biological samples in industrial areas. Wei et al.'s research on residents living in areas of heavily polluted by industry in China pointed out that arsenic in the air can be

absorbed by plants and animals through sedimentation, and then enter the human body through human consumption or drinking (Wei et al., 2017). In South Korea, a study revealed high concentrations of arsenic in water and soil near industrial areas, along with elevated levels in urine samples of nearby residents (Cho et al., 2013). Workplace investigations have also shown increased arsenic concentrations in urine samples of petrochemical workers in the Asaluye Petrochemical Industrial Zone within a two-week period (Kafaei et al., 2017). Therefore, it is known that arsenic in the petrochemical industrial area can be released into the environment in various forms, leading to environmental pollution, and subsequently entering the human body through inhalation, contact, and other routes. In Chan et al.'s researches at petrochemical industrial area at Taiwan in 2011 and 2013, the study areas were divided into three zones based on proximity to the petrochemical complex: within 10 km, 10–20 km, and 20–30 km away. They assessed air quality and arsenic levels in water samples. The concentration of arsenic in the particles collected at a short distance from the petrochemical industrial area is significantly higher than the non-affected ones when the wind is affected by the complex. Moreover, the arsenic content in drinking water near the petrochemical complex (1.12 ± 1.17 ng/ml) was significantly higher than in distant areas. The research also revealed higher arsenic content in the drinking water of residents near the petrochemical industrial area compared to other distant areas. Through environmental research and analysis of arsenic species in residents, it can be known that residents suffer from arsenic pollution. According to the health risk assessment project commissioned by the petrochemical complex in the present study between 2015 and 2020, the primary arsenic-containing feedstock utilized was arsenic trioxide. The emissions were predominantly attributed to coal-fired cogeneration plants. Total arsenic emissions within the petrochemical complex were recorded at 0.015 mt/yr in 2015, demonstrating a consistent annual decline to 0.007 mt/yr by 2020 (FPCC, 2025). Several studies have emphasized that the media and ways of inorganic arsenic exposure are very diverse. In addition to This study utilized local environmental testing and resident biopsy samples to examine exposure variations based on distance. It was also found through biological samples that the closer to the petrochemical industrial area, the higher the total arsenic concentration, and this difference is particularly pronounced in children. However, more studies and research samples are still needed to confirm the main exposure pathways and media of inorganic arsenic.

In order to assess the degree of methylation metabolism of inorganic arsenic, we calculated the values of PMI, SMI, (MMA + DMA)/iAs. PMI showed significant differences between the high and low exposure groups, regardless of the elderly or children group, indicating the poor conversion of iAs to MMA in the high-exposure group. A study on chronic arsenic exposure in Inner Mongolia pointed out that the methylation ability of arsenic decreased with the increase of arsenic exposure (Xu et al., 2008). Similarly, another study on occupational arsenic exposure in China reached a similar conclusion, suggesting that the main reason leading to the decrease in the methylation efficiency of arsenic species may be attributed to the consumption or inhibition of GSH, or the inhibition of S-adenosylmethionine (SAM), the methylation provider (Xi et al., 2011). Other studies have also indicated that exposure to arsenic pollution can reduce the methylation efficiency of arsenic and impair its detoxification effects, which are related to many diseases, such as cardiovascular disease, bladder cancer, increased oxidative stress, etc. (Tseng et al., 2005; Xu et al., 2008). As for the (MMA + DMA)/iAs, we only observed significant differences between the high and low exposure groups in the child group, while the elderly group could only be observed that the low-exposure group was higher than the high-exposure one. The value of (MMA + DMA)/iAs can be used to assess the level of primary methylation of arsenic, with higher values indicating better methylation efficiency. Based on the previous research data that the primary source of inorganic arsenic exposure is drinking water (Sun et al., 2007; Xu et al., 2008), it is consistent with our findings that the (MMA + DMA)/iAs value in the low-exposure group is higher than in the high-exposure group. Moreover, the (MMA + DMA)/iAs value in adults is already slightly lower than that in the child group under low-exposure or normal conditions, and there is no significant difference between the high-exposure groups of adults and children. Therefore, the decrease in the (MMA + DMA)/iAs value under high exposure appears to be more pronounced in the child group compared to adults, which is similar

to the results of our study. These findings show that the higher the concentration of inorganic arsenic in the body, the worse the methylation of arsenic. This effect is more prominent in children compared to the elderly. In addition, in our study, we observed that the SMI value in the children group was higher than in the elderly group, irrespective of exposure level, indicating that children have a stronger secondary methylation ability. In our study, the SMI value in the children group was higher than that in the elderly group, regardless of the exposure level, indicating that children have a stronger secondary methylation ability. This aligns with meta-analytic evidence indicating that older age was associated with higher percentage of MMA, lower percentage of DMA, and reduced secondary methylation indices (Shen et al., 2016). In contrast, previous studies showed that children tend to maintain higher percentage of DMA and lower percentage of iAs than adults (Skröder Löveborn et al., 2016). This greater metabolic plasticity in children was possibly due to the relatively preserved hepatic and renal function as well as shorter cumulative exposure to explain why the metabolite pattern of children was relatively unaffected by high exposure levels in other studies (Sun et al., 2007). The present study also found corresponding results that there was no significant difference in SMI between exposure levels in children despite their generally higher methylation capacity when compared to the elderly (Table 3).

We applied metabolomics to investigate the association between arsenic exposure and arsenic metabolism, and observed a significant decrease in urinary levels of glutamic acid, glycine, and methionine in the high exposure group compared to the low exposure group. Glutamic acid and glycine are important amino acids in the formation of glutathione. Previous studies have highlighted the importance of glutathione during inorganic arsenic metabolism. The antioxidant properties of glutathione will assist the transformation of As^{5+} to As^{3+}. However, inorganic arsenic can interfere with glutathione production and transformation, rendering it ineffective in metabolizing inorganic arsenic and leading to its accumulation in the body (Hall et al., 2013). On the other hand, methionine will transform into S-adenosylmethionine (SAM) in the body, serving as a vital methyl group provider in the one-carbon metabolism of inorganic arsenic. SAM is involved in the conversion of inorganic arsenic to organic arsenic and the transformation of MMA to DMA. Previous studies also showed that the deregulation of the methionine cycle associated with the development of cancer, cardiovascular diseases, and chronic kidney diseases (Locasale, 2013). These studies suggest that glutamic acid, glycine, and methionine will affect the metabolism of inorganic acid. Therefore, our discovery of reduced levels of glutamic acid, glycine, and methionine may provide a potential explanation for the higher levels of residual arsenic observed in our study subjects. Furthermore, we observed increased hippurate and serine in the high-exposure group compared with the low-exposure group. Previous research has shown that urinary hippurate increases with higher inorganic arsenic exposure, possibly due to the interactions between arsenic and thiol (–SH) groups that influence hippurate formation. Both hippurate and serine are known oxidative-stress-related metabolites linked to amino acid metabolism, which supports their elevation under arsenic-induced oxidative stress (Zhang et al., 2014). Previous studies showed that oxidative stress induced by arsenic exposure can cause DNA damage, playing an important role in the development of various diseases, including carcinogenesis, genotoxicity, diabetes, cardiovascular diseases, and nervous system diseases (Flora, 2011; Jomova et al., 2011). These observations suggest that the elevated exposure to inorganic acid could reduce arsenic metabolism, and further affecting people's health. More studies will be required for validation.

Limitation

There are several limitations to consider in this study. First, the arsenic species in biological samples cannot directly reflect the source of exposure. However, based on previous research on inorganic arsenic exposure in Taiwan, it can still be confirmed that the inorganic arsenic exposure in this area might be affected by the influence of nearby industrial areas. In addition, the study population is residents who have lived for years, and the presumed environmental pollution sources also exist stably for a long time. Therefore, although the half-life of arsenic in urine is not long, the content of arsenic species in our urine samples still can reflect the arsenic exposure and metabolism of the study subjects. Moreover, because this study employed a cross-sectional design, it cannot determine temporal ordering or establish a causal

relationship between arsenic exposure and metabolic alterations. Given that the nearby petrochemical complex has been operating for more than 20 years since 1999 and all participants have resided in the area for at least five years, long-term exposure to metal pollutants emitted from the facility is plausible.

Another limitation is the relatively small sample size. Although significant differences in arsenic species concentrations were still observed between the high- and low-exposure groups, the weak predictive performance in linking biological indicators with metabolite profiles may introduce bias and limit the representativeness of the findings. Increasing the sample size in future studies would enable a more comprehensive and sensitive evaluation of metabolite changes. Moreover, seafood is an important source of organic arsenic exposure. Although this study adjusted for seafood intake using questionnaire data, the assessment may not have been sufficiently detailed to fully account for its influence on arsenic speciation. Nevertheless, our study subjects, including both older adults and children, were comparable between the high and low exposure groups, as no significant differences in fish consumption were observed, suggesting that seafood intake is unlikely to serve as a potential confounder. Future studies should comprehensively assess dietary habits and measure metal levels in seafood to clarify its contribution to urinary arsenic levels.

Conclusion

Our study indicates that petrochemical industrial areas likely contribute to higher levels of arsenic exposure. Specifically, children were found to have higher exposure to both inorganic and organic arsenic, while the elderly showed higher exposure to inorganic arsenic. In addition, exposing to inorganic arsenic affects the expression of metabolites related to the methylation capacity of arsenic species and oxidative stress in vivo. Besides, the concentrations of inorganic arsenic in the children had a more pronounced effect on their primary methylation capacity compared to the elderly. Based on these findings, According to this study, we found that petrochemical industrial areas likely contribute to higher levels of arsenic exposure. Specifically, children were found to have higher exposure to both inorganic and organic arsenic, while the elderly showed higher exposure to inorganic arsenic. In addition, exposure to inorganic arsenic affects the expression of metabolites related to the methylation capacity of arsenic species and oxidative stress in vivo. Furthermore, the concentrations of inorganic arsenic in children had a more pronounced effect on their primary methylation capacity compared to the elderly. Based on these findings, we recommend continued monitoring of inorganic arsenic exposure among local residents, including measurements of arsenic in drinking water and ambient air, biomonitoring of urinary arsenic species, and routine health assessments to detect early adverse effects. Furthermore, longitudinal cohort studies are warranted to evaluate the long-term health impacts of chronic arsenic exposure. It is important to pay particular attention to the inorganic arsenic exposure and arsenic metabolism of local individuals, especially children, in areas with higher exposure. Efforts should be made to minimize the possibility of exposure to inorganic arsenic as much as possible. It is important to pay particular attention to the inorganic arsenic exposure and arsenic metabolism of local individuals, especially children, in areas with higher exposure. Efforts should be made to minimize the possibility of exposure to inorganic arsenic as much as possible. In addition, some policy and management were recommended to reduce arsenic exposure risks for children and the elderly living in the vicinity of petrochemical industry. Potential measures include decommissioning aging coal-fired units, implementing stringent emission controls, temporarily shutting down high-emission units during peak pollution periods, and strengthening the pollution management and control in key facilities or areas.

Acknowledgements This study was funded with a grant obtained from the Public Health Bureau, Yunlin County, Taiwan (YLPHB-10512). We also thank the funding from The Ministry of Science and Technology, Taiwan (grant number MOST 107-3017-F-002-003, MOST 108-3017-F-002-001 and MOST 104-2621-M-002-012-MY3).

Author contributions Tzu-Hsuen Yuan: Conceptualization, Investigation, Methodology, Project administration, Validation, Writing—original draft, Writing—review & editing. Yi-Ting Hsieh: Data curation, Formal analysis, Project administration, Software, Validation, Visualization, Writing—original draft. Chi-Hsin Chen: Data curation, Formal analysis, Methodology,

Software, Validation. Chia-Yu Chang: Data curation, Validation, Visualization, Writing—review & editing. Chang-Chuan Chan: Conceptualization, Funding acquisition, Resources, Supervision, Validation, Writing—review & editing.

Funding This work was supported by the Public Health Bureau, Yunlin County, Taiwan (grant numbers: YLPHB-10512); The Ministry of Science and Technology, Taiwan (grant number MOST 107–3017-F-002–003, MOST 108–3017-F-002–001 and MOST 104–2621-M-002–012-MY3).

Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Ethical approval and consent to participate This study received approval from the Research Ethics Committee of the National Health Research Institutes in Taiwan (approval number: 201505016RINB) and was conducted in accordance with the Declaration of Helsinki. Consent from subjects participating in the study was received prior to conducting the study.

References

Bosco, M. L., Varrica, D., & Dongarrà, G. (2005). Case study: Inorganic pollutants associated with particulate matter from an area near a petrochemical plant. *Environmental Research, 99*(1), 18–30. https://doi.org/10.1016/j.envres.2004.09.011

Byun, K., Won, Y., Hwang, Y., Koh, D.-H., Im, H., & Kim, E.-A. (2013). Assessment of arsenic exposure by measurement of urinary speciated inorganic arsenic metabolites in workers in a semiconductor manufacturing plant. *Annals of Occupational and Environmental Medicine*. https://doi.org/10.1186/2052-4374-25-21

Chan, C. C., Lee, Y. L., & Hung, S. H. (2012). Air pollutants and health among residents near a petrochemical complex in Yunlin Country : a cohort study (Project No. YLEPB-100–029). Taiwan Yunlin County Health Bureau.

Chan, C. C., Wang, C. L., Chen, Y. M., Chen, C. Y., & Tsai, C. H. (2015). Environmental epidemiology cohort study in Yunlin County (Project No. 10216). Taiwan: Yunlin County Health Bureau.

Chang, J. Y., Ahn, S. C., Lee, J. S., Kim, J. Y., Jung, A. R., Park, J., Choi, J. W., & Do Yu, S. (2019). Exposure assessment for the abandoned metal mine area contaminated by arsenic. *Environmental Geochemistry and Health, 41*(6), 2443–2458. https://doi.org/10.1007/s10653-019-00296-5

Chen, C. S., Kuo, T. C., Kuo, H. C., Tseng, Y. J., Kuo, C. H., Yuan, T. H., & Chan, C. C. (2019). Metabolomics of children and adolescents exposed to industrial carcinogenic pollutants. *Environmental Science & Technology, 53*(9), 5454–5465. https://doi.org/10.1021/acs.est.9b00392

Chen, C. S., Yuan, T. H., Shie, R. H., Wu, K. Y., & Chan, C. C. (2017). Linking sources to early effects by profiling urine metabolome of residents living near oil refineries and coal-fired power plants. *Environment International, 102*, 87–96. https://doi.org/10.1016/j.envint.2017.02.003

Chen, Y. C., Hsu, C. Y., Lin, S. L., et al. (2015). Characteristics of concentrations and metal compositions for $PM_{2.5}$ and $PM_{2.5–10}$ in Yunlin County, Taiwan during air quality deterioration. *Aerosol and Air Quality Research, 15*, 2571–2583. https://doi.org/10.4209/aaqr.2015.04.0261

Cho, Y., Seo, S., Choi, S. H., Lee, S., Kim, K., Kim, H. J., & Choi, J. W. (2013). Association of arsenic levels in soil and water with urinary arsenic concentration of residents in the vicinity of closed metal mines. *International Journal of Hygiene and Environmental Health, 216*(3), 255–262. https://doi.org/10.1016/j.ijheh.2012.05.003

European Commission, Directorate-General for Environment. (n.d.). EU air quality standards. European Commission. https://environment.ec.europa.eu/topics/air/air-quality/eu-air-quality-standards_en

Flora, S. J. (2011). Arsenic-induced oxidative stress and its reversibility. *Free Radical Biology & Medicine, 51*(2), 257–281. https://doi.org/10.1016/j.freeradbiomed.2011.04.008

Formosa Petrochemical Corporation (FPCC), 2025. Health Risk Assessment. https://crm.fpg.com.tw/j2sha/cus/art/Cc1d04.do (accessed 24 November 2025)

Gamble, M. V., Liu, X., Ahsan, H., Pilsner, R., Ilievski, V., Slavkovich, V., Parvez, F., Levy, D., Factor-Litvak, P., & Graziano, J. H. (2005). Folate, homocysteine, and arsenic metabolism in arsenic-exposed individuals in Bangladesh. *Environmental Health Perspectives, 113*(12), 1683–1688. https://doi.org/10.1289/ehp.8084

Giri, B., & Dey, S. (2017). Is it possible to avert arsenic effects on cells and tissues bypassing its toxicity and suppressive consequences of energy production? A hypothesis. *BLDE University Journal of Health Sciences, 2*, 91–96.

Hall, M. N., Niedzwiecki, M., Liu, X., Harper, K. N., Alam, S., Slavkovich, V., Ilievski, V., Levy, D., Siddique, A. B., Parvez, F., Mey, J. L., van Geen, A., Graziano, J., & Gamble, M. V. (2013). Chronic arsenic exposure and blood glutathione and glutathione disulfide concentrations in Bangladeshi adults. *Environmental Health Perspectives, 121*(9), 1068–1074.

Hastings, J., De Matos, P., Dekker, A., Ennis, M., Harsha, B., Kale, N., et al. (2012). The ChEBI reference database and ontology for biologically relevant chemistry: Enhancements for 2013. *Nucleic Acids Research, 41*(D1), D456–D463. https://doi.org/10.1093/nar/gks1146

Hong, Y. S., Song, K. H., & Chung, J. Y. (2014). Health effects of chronic arsenic exposure. *Journal of Preventive Medicine and Public Health, 47*(5), 245–252. https://doi.org/10.3961/jpmph.14.035

Hsueh, Y. M., Chung, C. J., Shiue, H. S., Chen, J. B., Chiang, S. S., Yang, M. H., Tai, C. W., & Su, C. T. (2009). Urinary arsenic species and CKD in a Taiwanese population: A case-control study. *American Journal of Kidney Diseases, 54*(5), 859–870. https://doi.org/10.1053/j.ajkd.2009.06.016

Jennrich, P. (2013). The influence of arsenic, lead, and mercury on the development of cardiovascular diseases. *ISRN Hypertension, 2013*, Article 234034. https://doi.org/10.5402/2013/234034

Jiang, S., Su, J., Yao, S., Zhang, Y., Cao, F., Wang, F., Wang, H., Li, J., & Xi, S. (2014). Fluoride and arsenic exposure impairs learning and memory and decreases mGluR5 expression in the hippocampus and cortex in rats. *PLoS ONE, 9*(4), Article e96041. https://doi.org/10.1371/journal.pone.0096041

Jomova, K., Jenisova, Z., Feszterova, M., Baros, S., Liska, J., Hudecova, D., Rhodes, C. J., & Valko, M. (2011). Arsenic: Toxicity, oxidative stress and human disease. *Journal of Applied Toxicology, 31*(2), 95–107. https://doi.org/10.1002/jat.1649

Kafaei, R., Tahmasbi, R., Ravanipour, M., Vakilabadi, D. R., Ahmadi, M., Omrani, A., & Ramavandi, B. (2017). Urinary arsenic, cadmium, manganese, nickel, and vanadium levels of schoolchildren in the vicinity of the industrialised area of Asaluyeh, Iran. *Environmental Science and Pollution Research International, 24*(30), 23498–23507. https://doi.org/10.1007/s11356-017-9981-6

Kanehisa, M., Goto, S., Sato, Y., Kawashima, M., Furumichi, M., & Tanabe, M. (2014). Data, information, knowledge and principle: back to metabolism in KEGG. *Nucleic Acids Research, 42*(D1), D199–D205. https://doi.org/10.1093/nar/gkt1076

Kuo, C. C., Moon, K. A., Wang, S. L., Silbergeld, E., & Navas-Acien, A. (2017). The association of arsenic metabolism with cancer, cardiovascular disease, and diabetes: A systematic review of the epidemiological evidence. *Environmental Health Perspectives, 125*(8), Article 087001. https://doi.org/10.1289/ehp577

Lasky, T., Sun, W., Kadry, A., & Hoffman, M. K. (2004). Mean total arsenic concentrations in chicken 1989-2000 and estimated exposures for consumers of chicken. *Environmental Health Perspectives, 112*(1), 18–21. https://doi.org/10.1289/ehp.6407

Lawton, K. A., Berger, A., Mitchell, M., Milgram, K. E., Evans, A. M., Guo, L., Hanson, R. W., Kalhan, S. C., Ryals, J. A., & Milburn, M. V. (2008). Analysis of the adult human plasma metabolome. *Pharmacogenomics, 9*(4), 383–397. https://doi.org/10.2217/4622416.9.4.383

Li, H., Wang, M., Liang, Q., Jin, S., Sun, X., Jiang, Y., Pan, X., Zhou, Y., Peng, Y., Zhang, B., Zhou, A., Zhang, Y., Chen, Z., Cao, J., Zhang, H., Xia, W., Zheng, T., Cai, Z., Li, Y., & Xu, S. (2017). Urinary metabolomics revealed arsenic exposure related to metabolic alterations in general Chinese pregnant women. *Journal of Chromatography A, 1479*, 145–152. https://doi.org/10.1016/j.chroma.2016.12.007

Locasale, J. W. (2013). Serine, glycine and one-carbon units: Cancer metabolism in full circle. *Nature Reviews. Cancer, 13*(8), 572–583. https://doi.org/10.1038/nrc3557

Navas-Acien, A., Francesconi, K. A., Silbergeld, E. K., & Guallar, E. (2011). Seafood intake and urine concentrations of total arsenic, dimethylarsinate and arsenobetaine in the US population. *Environmental Research, 111*(1), 110–118. https://doi.org/10.1016/j.envres.2010.10.009

Patel, K. S., Shrivas, K., Brandt, R., Jakubowski, N., Corns, W., & Hoffmann, P. (2005). Arsenic contamination in water, soil, sediment and rice of central India. *Environmental Geochemistry and Health, 27*(2), 131–145. https://doi.org/10.1007/s10653-005-0120-9

Petrick, J. S., Ayala-Fierro, F., Cullen, W. R., Carter, D. E., & Vasken Aposhian, H. (2000). Monomethylarsonous acid (MMA(III)) is more toxic than arsenite in Chang human hepatocytes. *Toxicology and Applied Pharmacology, 163*(2), 203–207. https://doi.org/10.1006/taap.1999.8872

Rahman, M., Vahter, M., Sohel, N., Yunus, M., Wahed, M. A., Streatfield, P. K., Ekström, E.-C., & Persson, L. Å. (2006). Arsenic exposure and age- and sex-specific risk for skin lesions: A population-based case–referent study in Bangladesh. *Environmental Health Perspectives, 114*(12), 1847–1852. https://doi.org/10.1289/ehp.9207

Ratnaike, R. N. (2003). Acute and chronic arsenic toxicity. *Postgraduate Medical Journal, 79*(933), 391–396. https://doi.org/10.1136/pmj.79.933.391

Shen, H., Niu, Q., Xu, M., Rui, D., Xu, S., Feng, G., Ding, Y., Li, S., & Jing, M. (2016). Factors affecting arsenic methylation in arsenic-exposed humans: A systematic review and meta-analysis. *International Journal of Environmental Research and Public Health, 13*(2), 205. https://doi.org/10.3390/ijerph13020205

Skröder Löveborn, H., Kippler, M., Lu, Y., Ahmed, S., Kuehnelt, D., Raqib, R., & Vahter, M. (2016). Arsenic Metabolism in Children Differs From That in Adults. *Toxicological Sciences : an Official Journal of the Society of Toxicology*. https://doi.org/10.1093/toxsci/kfw060

Smedley, P. L., & Kinniburgh, D. G. (2002). A review of the source, behaviour and distribution of arsenic in natural waters. *Applied Geochemistry, 17*(5), 517–568. https://doi.org/10.1016/S0883-2927(02)00018-5

Sun, G., Xu, Y., Li, X., Jin, Y., Li, B., & Sun, X. (2007). Urinary arsenic metabolites in children and adults exposed to arsenic in drinking water in Inner Mongolia China. *Environmental Health Perspectives, 115*(4), 648–652. https://doi.org/10.1289/ehp.9271

Taylor, V., Goodale, B., Raab, A., Schwerdtle, T., Reimer, K., Conklin, S., Karagas, M. R., & Francesconi, K. A. (2017). Human exposure to organic arsenic species from seafood. *Science of the Total Environment, 580*, 266–282. https://doi.org/10.1016/j.scitotenv.2016.12.113

Tseng, C. H., Huang, Y. K., Huang, Y. L., Chung, C. J., Yang, M. H., Chen, C. J., & Hsueh, Y. M. (2005). Arsenic exposure, urinary arsenic speciation, and peripheral vascular disease in blackfoot disease-hyperendemic villages in Taiwan. *Toxicology and Applied Pharmacology, 206*(3), 299–308. https://doi.org/10.1016/j.taap.2004.11.022

Tyler, C. R., & Allan, A. M. (2014). The effects of arsenic exposure on neurological and cognitive dysfunction in human and rodent studies: A review. *Current Environmental Health Reports, 1*(2), 132–147. https://doi.org/10.1007/s40572-014-0012-1

Wang, C. W., Liao, K. W., Chan, C. C., Yu, M. L., Chuang, H. Y., Chiang, H. C., & Huang, P. C. (2019). Association between urinary thiodiglycolic acid level and hepatic function or fibrosis index in school-aged children living near a petrochemical complex. *Environmental Pollution (Barking, Essex : 1987), 244*, 648–656. https://doi.org/10.1016/j.envpol.2018.10.012

Wang, S.-L., Chang, F.-H., Liou, S.-H., Wang, H.-J., Li, W.-F., & Hsieh, D. P. H. (2007). Inorganic arsenic exposure and its relation to metabolic syndrome in an industrial area of Taiwan. *Environment International, 33*(6), 805–811. https://doi.org/10.1016/j.envint.2007.03.004

Wei, B., Yu, J., Kong, C., Li, H., Yang, L., Guo, Z., Cui, N., Xia, Y., & Wu, K. (2017). An investigation of the health effects caused by exposure to arsenic from drinking water and coal combustion: Arsenic exposure and metabolism. *Environmental Science and Pollution Research, 24*(33), 25947–25954. https://doi.org/10.1007/s11356-017-0203-z

Weng, H. H., Tsai, S. S., Chiu, H. F., Wu, T. N., & Yang, C. Y. (2008). Association of childhood leukemia with residential exposure to petrochemical air pollution in Taiwan. *Inhalation Toxicology, 20*(1), 31–36. https://doi.org/10.1080/08958370701758734

Wishart, D. S., Feunang, Y. D., Marcu, A., Guo, A. C., Liang, K., Vázquez-Fresno, R., et al. (2018). HMDB 4.0: the human metabolome database for 2018. *Nucleic Acids Research, 46*(D1), D608–D617. https://doi.org/10.1093/nar/gkx1089

World Health Organization. (2022). Guidelines for drinking-water quality: Fourth edition incorporating the first and second addenda. World Health Organization.

Xi, S., Zheng, Q., Zhang, Q., & Sun, G. (2011). Metabolic profile and assessment of occupational arsenic exposure in copper- and steel-smelting workers in China. *International Archives of Occupational and Environmental Health, 84*(3), 347–353. https://doi.org/10.1007/s00420-010-0574-7

Xu, Y., Wang, Y., Zheng, Q., Li, X., Li, B., Jin, Y., Sun, X., & Sun, G. (2008). Association of oxidative stress with arsenic methylation in chronic arsenic-exposed children and adults. *Toxicology and Applied Pharmacology, 232*(1), 142–149. https://doi.org/10.1016/j.taap.2008.06.010

Yamauchi, H., Takahashi, K., Mashiko, M., & Yamamura, Y. (1989). Biological monitoring of arsenic exposure of gallium arsenide- and inorganic arsenic-exposed workers by determination of inorganic arsenic and its metabolites in urine and hair. *American Industrial Hygiene Association Journal, 50*(11), 606–612. https://doi.org/10.1080/15298668991375236

Yuan, T. H., Chio, C. P., Shie, R. H., Pien, W. H., & Chan, C. C. (2016). The distance-to-source trend in vanadium and arsenic exposures for residents living near a petrochemical complex. *Journal of Exposure Science & Environmental Epidemiology, 26*(3), 270–276. https://doi.org/10.1038/jes.2015.2

Yuan, T. H., Shen, Y. C., Shie, R. H., Hung, S. H., Chen, C. F., & Chan, C. C. (2018). Increased cancers among residents living in the neighborhood of a petrochemical complex: a 12-year retrospective cohort study. *International Journal of Hygiene and Environmental Health, 221*(2), 308–314. https://doi.org/10.1016/j.ijheh.2017.12.004

Yuan, T. H., Ke, D. Y., Wang, J. E. H., & Chan, C. C. (2020). Associations between renal functions and exposure of arsenic and polycyclic aromatic hydrocarbon in adults living near a petrochemical complex. *Environmental Pollution, 256*, Article 113457. https://doi.org/10.1016/j.envpol.2019.113457

Zhang, J., Shen, H., Xu, W., Xia, Y., Barr, D. B., Mu, X., Wang, X., Liu, L., Huang, Q., & Tian, M. (2014). Urinary metabolomics revealed arsenic internal dose-related metabolic alterations: A proof-of-concept study in a Chinese male cohort. *Environmental Science & Technology, 48*(20), 12265–12274. https://doi.org/10.1021/es503659w

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.